

PROJECT AND TEAM INFORMATION

Project Title

Image Captioning Using Machine Learning: A CNN (InceptionV3)–LSTM Encoder–Decoder Approach

Student / Team Information

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Motivation (1 pt)

Every day, billions of images are uploaded across social media, e-commerce platforms, and surveillance systems, and manually describing this volume of visual content is simply not feasible. Automatically generating natural-language descriptions of images has wide practical value: it powers assistive technology that helps visually impaired users understand visual content, improves searchability and indexing of large image repositories, and supports applications in e-commerce, robotics, and surveillance. Beyond its practical use, image captioning is academically compelling because it sits at the intersection of two challenging AI domains — computer vision, which must understand what is present in an image, and natural language processing, which must express that understanding as a coherent, grammatically correct sentence. This project builds a complete image captioning pipeline using a pretrained InceptionV3 convolutional network as an image encoder and an LSTM network as a language decoder, and explores transfer learning as a practical way to achieve strong results without training a vision model from scratch.

State of the Art / Current solution (1 pt)

Image captioning has evolved through several generations of approaches. Early systems relied on template-based or retrieval-based methods that matched images to a fixed bank of captions and produced rigid, non-fluent output. The current standard approach, and the one this project follows, is the encoder–decoder architecture: a pretrained convolutional neural network (such as InceptionV3, VGG16, or ResNet) extracts a fixed-length feature vector from the image, and a recurrent decoder (typically an LSTM or GRU) generates the caption word by word conditioned on those features. More recent state-of-the-art systems extend this with attention mechanisms (Bahdanau or Luong attention) that let the decoder focus on different image regions as it generates each word, and the newest generation uses transformer-based architectures such as ViT-GPT2, BLIP, and ClipCap, which achieve higher fluency and accuracy but require significantly more training data and compute. This project implements the classical CNN–LSTM encoder–decoder baseline, which remains a strong and computationally practical starting point.

Project Goals and Milestones (2 pts)

Develop a deep learning model that generates accurate, meaningful captions for input images. Use a pretrained InceptionV3 network for efficient visual feature extraction (transfer learning). Design and train an LSTM-based decoder that generates grammatically correct captions.

Milestone 1: Acquire and preprocess the Kaggle/Flickr8k dataset — clean captions, tokenize text, build vocabulary, and normalize/resize images to 299×299 .

Milestone 2: Extract InceptionV3 image features and build the merged encoder–decoder training pipeline.

Milestone 3: Train the model using categorical cross-entropy loss and the Adam optimizer, and tune hyperparameters.

Milestone 4: Evaluate the trained model using BLEU, METEOR, ROUGE, and CIDEr scores.

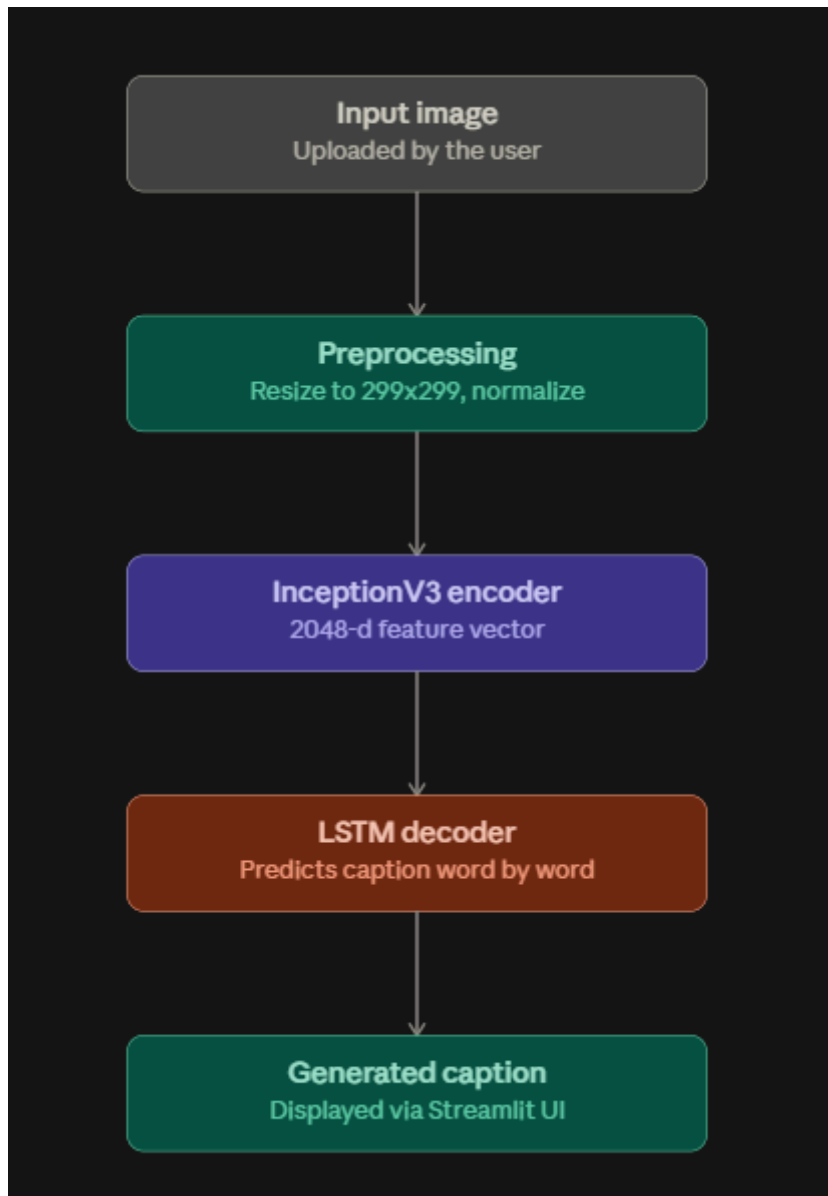
Milestone 5: Deploy the final trained model behind a user-friendly Streamlit web interface for real-time caption generation.

Project Approach (3 pts)

The project follows a standard encoder–decoder workflow. First, a cleaned and structured image–caption dataset (Kaggle / Flickr8k) is prepared. Images are resized to 299×299 and pixel values are normalized before being passed through a pretrained InceptionV3 network, which produces a 2048-dimensional feature vector per image; captions are cleaned, tokenized, converted into a vocabulary, and padded into fixed-length sequences. These visual and textual features are combined in a merged encoder–decoder model, where the LSTM decoder predicts the caption one word at a time using the image features together with previously generated words, continuing until an end-of-sequence token is produced. The model is trained with categorical cross-entropy loss and the Adam optimizer, and captions are generated at inference time using either greedy search or beam search.

Technology stack: Python, TensorFlow/Keras for model development, InceptionV3 (pretrained, ImageNet weights) as the CNN encoder, an LSTM network as the decoder, and Streamlit for deploying a real-time, browser-based captioning interface.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The primary deliverable is a trained CNN–LSTM image captioning model, packaged with its preprocessing pipeline and evaluated using BLEU-1 through BLEU-4, METEOR, ROUGE, and CIDEr scores against held-out validation data. Alongside the model, the project delivers a functional Streamlit web application that lets a user upload an image and receive a generated caption in under five seconds, together with supporting test suites (unit, integration, performance, and error-handling tests) and full project documentation, including training/validation loss curves and sample qualitative results (e.g., an image of a person riding a horse correctly captioned as "A man riding a horse.").

Assumptions

It is assumed that the Kaggle/Flickr8k dataset provides sufficient image–caption pairs to train a reasonably generalizing model within the project timeline; that pretrained InceptionV3 (ImageNet) weights are available and suitable for transfer learning without additional licensing constraints; that adequate GPU compute is available for training within a reasonable time; and that the deployed Streamlit application will be used locally/for demonstration purposes rather than at production scale.

References

Flickr8k / Kaggle Image Captioning Dataset — <https://www.kaggle.com/datasets/adityajn105/flickr8k>
Streamlit Documentation — <https://docs.streamlit.io>
TensorFlow / Keras Documentation — <https://www.tensorflow.org>